

Chapter 31

Video and Audio Effects in the Cut Page

As you build your edit, you'll often find it necessary to add effects to clips to create titles, blend clips together with compositing, add plugins to give clips a particular look, speed clips up or slow them down, or transform clips to zoom into them or move them around the frame. The Cut page has controls to accomplish all of this and more.

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Adding Transitions

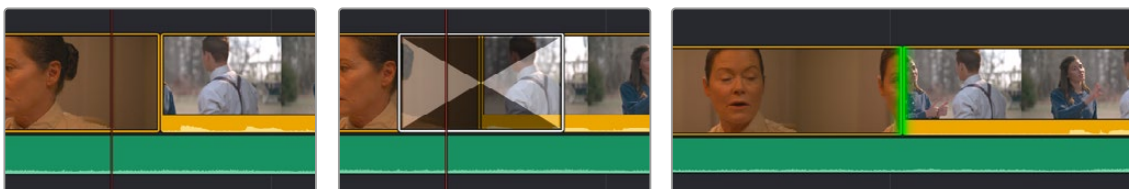
You can add transitions such as dissolves or wipes to an edit to indicate a change of topic, a change of location, or the passage of time. You can also add a Smooth Cut transition to patch unwanted cuts in interview clips or other situations where there's not a significant change in the position of the subject or background in the frame. Three buttons, at the bottom right of the Media Pool, make it easy to add and remove dissolves and Smooth Cut transitions.



The Cut, Dissolve, and Smooth Cut buttons

Adding Dissolves

Dissolves are the most standard and ubiquitous type of transition, and so they're the easiest to create. Place the playhead on or near an edit you want to turn into a dissolve, and click the Dissolve button at the bottom left of the Media Pool. A one second Cross Dissolve will be added to the edit point that's nearest to the edit point.

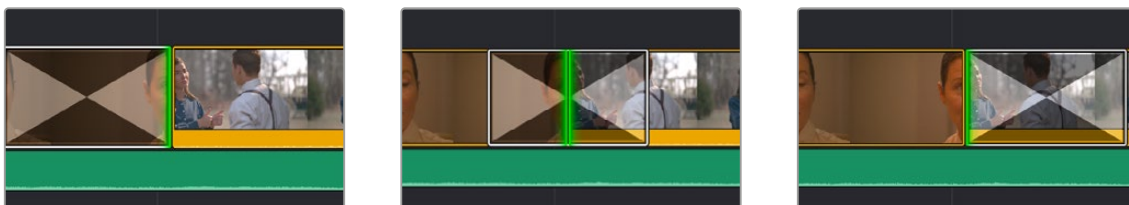


(Left) Moving a playhead near an edit point, (Middle) Adding a dissolve (Right) A selected edit point

Adding Dissolve Using Keyboard Shortcuts

You can also add a dissolve by selecting one or more edit points (or moving the playhead near an edit you want to select and pressing V to select the nearest edit point), then pressing Command-T to add a dissolve, which is the standard transition.

You can also add a transition that's aligned with the beginning, middle, or end of the edit point using the keyboard. To do so, select an edit point, press the U key repeatedly to cycle among selecting the start, center, or end of the edit, then press Command-T to add the standard transition. The standard transition will be added with its alignment based on the edit selection you made; selecting the start of the edit places a transition that ends on the edit; selecting the end of the edit places a transition that starts on the edit, and choosing the center of the edit places a transition that is similarly centered.



A transition that's aligned with the beginning, center, and end of an edit point

Changing a Transition to a Cut

To remove a dissolve, thereby changing a transition to a cut, move the playhead at or near the edit with the dissolve, and click the Cut button, which is fast to do when you're using the DaVinci Resolve Editor Keyboard which has a dedicated Cut button.

TIP: Alternately, you can select one or more transitions in the Timeline and press the Delete key to remove them.

Adding Smooth Cuts

Smooth Cuts are special-purpose transitions designed to make short jump cuts in the middle of a clip unnoticeable. This is done by using AI powered Speed Warp processing to match the same features on either side of a cut in order to automatically morph a subject from one position to another over the duration of the transition.

To add a Smooth Cut:

- 1 Place the playhead on or near an edit.
- 2 Click the Smooth Cut button at the bottom left of the Media Pool. A one second Smooth Cut will be added to the edit point that's nearest to the edit point.

A Mode dropdown menu provides two options: Faster and Better. The Better option is default, with excellent quality and the capability of preserving the motion of subjects for the duration of the transition. The Faster option morphs between stills of the outgoing and incoming frames. In most practical circumstances, the Better mode will give you a superior result, but certain cuts or effects may be better addressed with the Faster option.

The Smooth Cut effect works best on clips such as sit-down interviews and close-up head shots with a minimum of background and subject motion, and where the subject's position on either side of the cut is not significantly different. A good example of when Smooth Cut is effective is when you're cutting pauses, partial repeats, filler sounds such as "um" or "you know," or other speech disfluencies out of an interview clip to tighten the dialog, and you want to eliminate the little "jump" that occurs at the cut without having to cut away to B-roll. Applying a short two or four frame Smooth Cut transition to the edit can make this kind of edit invisible, as long as the speaker doesn't change position significantly during the cut. The more motion there is in the background of the shot, and the more the speaker changes position, the harder it will be to get a useful result using Smooth Cut. Although the default duration for any transition is one second, you'll find that Smooth Cut transitions may work much better when they're short; 2- to 6-frame Smooth Cut transitions often work best to disguise jump cuts.

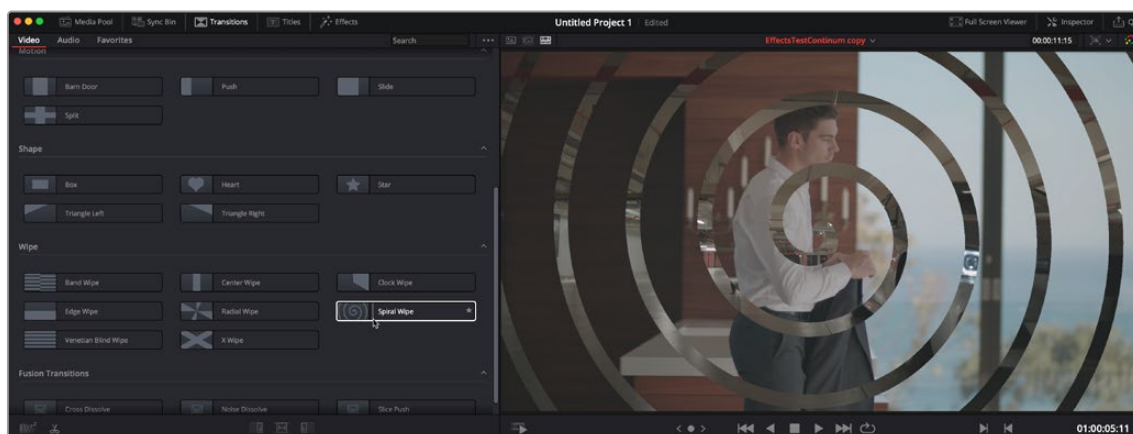
NOTE: As of DaVinci Resolve 20, Smooth Cut uses a new AI powered Speed Warp to perform the transition. If you would prefer to revert to the older optical flow version, simply uncheck the Speed Warp box in the Transition inspector.

Adding Other Kinds of Transitions

In order to make the selection of transitions, titles, and effects more intuitive, DaVinci Resolve shows each effect as a thumbnail representation in addition to the text name. This allows the user to quickly scan through all the numerous options to select the appropriate effect visually, rather than remembering them based on name alone. You can preview transitions and titles before you place them on the Timeline to quickly audition multiple options before making your final decision.

Transitions Thumbnails

To preview a transition before you place it into the Timeline, ensure that "Hover Scrub Preview" is checked in the Transitions option menu, then simply hover your pointer over any transition in the Transitions tab and move it across the thumbnail. The transition will preview in the Viewer using the two clips nearest the Smart Indicator in the Cut page, or the two clips nearest the playhead in the Edit page.



Scrubbing over a Transition Thumbnail previews the transition in the Viewer.

Once you've chosen your transition, it can be applied by to any edit point by using the methods below:

Methods of adding different transitions:

- **To add a transition by dragging it from the Transitions Browser:** Drag a video transition from the Effects Library to an edit point in the Timeline so that it's centered at, ends at, or starts at the edit point. If there is no overlap between the heads and tails of the two clips, you may not be able to add a transition where you want.
- **To add a transition using the Transitions Browser's contextual menu:** Select one or more edit points (one per track), then right-click a video transition in the Effects Library and choose Add to Selected Edit Points and Clips. That transition will be added to every selected edit point at once.
- **To add a transition by double-clicking in the Transition Browser:** You can double-click a transition in the Cut page's Transitions Browser to apply it directly to the edit point referenced by the Smart Indicator.
- **To add a transition by using the transition alignment icons in the Transition Browser:** You can double-click on one of three transition alignment icons at the bottom of the in the Cut page's Transition Browser.
- **To add a transition to all edit points on a timeline:** Click on the Timeline Actions menu and select Add Transition to All Edits. This will add the last used transition to all the edit points on a timeline.

Favorited Transitions

Any transitions you've marked as favorites (either by clicking on the star icon on its thumbnail or by selecting Add to Favorites from the contextual menu) can be accessed in the Favorites tab of the Effects library.

Editing and Removing Transitions

Once you've added a transition, you can edit it in a number of different ways to work best for your program, both in the Timeline and in the Transition Inspector.

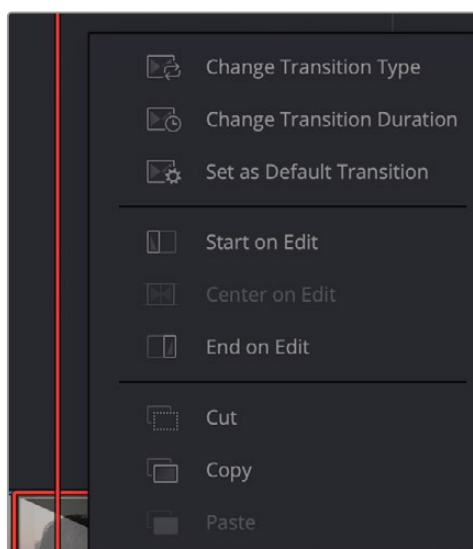
Methods of editing transitions in the Timeline:

- **To change a transition's duration:** Drag the beginning or end of a transition in the Timeline to be longer or shorter symmetrically about the current edit.
- **To move a transition from one edit to another:** You can drag a transition from its current position to another edit point.

- **To copy a transition from one edit to another:** Select a transition, then option-drag it to another edit point to add a duplicate transition.
- **To change a transition's type:** Drag a different transition from the Transition Browser onto the current one in the Timeline.
- **To remove a transition:** Select a transition in the Timeline and press the Delete key.
- **To remove all transitions in a timeline:** Click on the Timeline Actions menu and select Clear Transition from All Edits.

Editing transitions in the Transition Contextual menu:

You can edit the parameters of your transition, directly in the Timeline, by right-clicking on the transition and selecting one of the options from the contextual menu.



Right-clicking on a transition in the Timeline gives you the editing options above.

- **Change Transition Type:** You can change to any other installed transition by selecting one in the drop-down menu.
- **Change Duration:** Set the duration of the transition, shown in both seconds and frames.
- **Set as Default Transition:** Sets the currently selected transition as the Default Transition.
- **Alignment Options:** Lets you choose the transition's position relative to the edit point it's applied to. Your choices are "Start on Edit," "Center on Edit," and "End on Edit."
- **Cut/Copy/Paste:** Standard cut, copy, and paste controls for editing transitions in the timeline.

Editing transitions in the Transition Inspector:

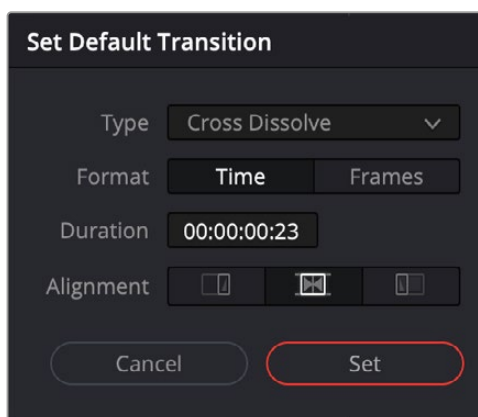
For more detailed control over your transition, including modifying all transition-specific parameters, use the Transition Inspector.

For more information on each specific transition included in DaVinci Resolve, see *Chapter 30, "Using the Inspector in the Cut Page."*

Setting the Default Transition

By default, the default transition is a simple cross dissolve. If you wish to change that transition to another one, simply right-click on a transition thumbnail and select Set as Standard Transition from the contextual menu.

You not only change the Default Transition type but also its duration and alignment from either the Timeline Options menu, or by right-clicking on the transition in the Timeline and selecting “Set Default Transition.”



Changing the Default Transition

Titles

There's a collection of title clips in the Titles Browser that you can use to add superimposed titles, slates, interstitial titles, lower thirds, or otherwise fulfill any textual needs your program has.

Titles Thumbnails

To audition titles before you place them into the Timeline, ensure that “Hover Scrub Preview” is checked in the Titles option menu, then simply hover your pointer over any thumbnail in the Titles tab. If the title is animated (i.e., Fusion titles), moving the pointer across the thumbnail will preview the animation. Once you've chosen your title, you can drag it from the Titles tab to your Timeline in the Edit page or in the Cut page to either the upper or lower Timelines, or use the editing selection modes at the bottom of the tab.



Scrubbing over a Title Thumbnail previews the title in the Viewer.

Two categories present two different kinds of titles. The “Titles” category presents simple, bare-bones titles that you can customize in a variety of different ways. Of the available options, the Text title is the most flexible. The “Fusion Titles” category presents more complicated titles that are more visually sophisticated and have more preset animation.

Adding Titles

You can add a title to the Timeline by opening the Titles Browser and dragging and dropping titles into the Timeline wherever you want them to be, just like any other clip. The default duration of a title clip is 5 seconds. This standard “generator” duration can be customized in the Editing panel of the User Preferences.

Once edited into the Timeline, titles can be moved, resized, and superimposed much like any other clip. Furthermore, when selected, both titles and generators expose the same Transform, Cropping, Speed, and Composite controls as any other clip; these controls can be used to composite titles in various ways to create different text effects.

Editing Titles

Once you add a title to the Timeline, the original title generators that shipped with DaVinci Resolve have onscreen controls that let you edit text and transform and position blocks of text directly within the Timeline Viewer.

Positioning and Transforming Text

So long as the Timeline playhead is positioned over a text generator that’s on top of one or more background clips, clicking on the text in the Timeline Viewer reveals onscreen transform controls that correspond to the Position, Zoom, and Rotation parameters in the Inspector.



Transforming a title’s text box in the Viewer

While dragging text to reposition it, snapping occurs at the X and Y center of the frame, as well as around the outer third of the frame. Holding the Shift key down while dragging a text object constrains movement to just the X or Y axes. Holding the Option key down turns snapping off.

Editing Text

Double-clicking on text in the Timeline Viewer puts that text into an editable state, wherein you can insert a text cursor or select characters to edit the text as you would in any text editor.

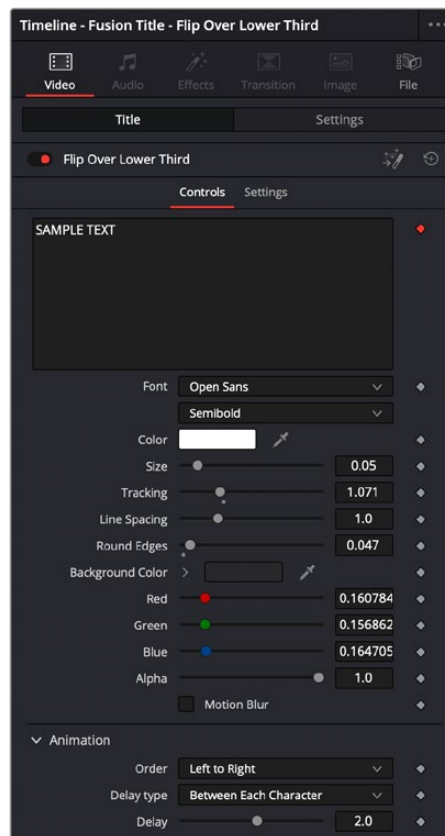


Editing the text of a title in the Viewer

Editing Titles Using the Video Inspector

For more precise control of your titles, double-clicking on any title will bring up its parameters in the Video Inspector. Here you can change fonts, colors, drop shadows, backgrounds, etc. You can also adjust basic transform parameters and cropping as well.

For more information on Editing Titles, see *Chapter 49, “Titles, Generators, and Stills.”*



The Title settings of Video Inspector

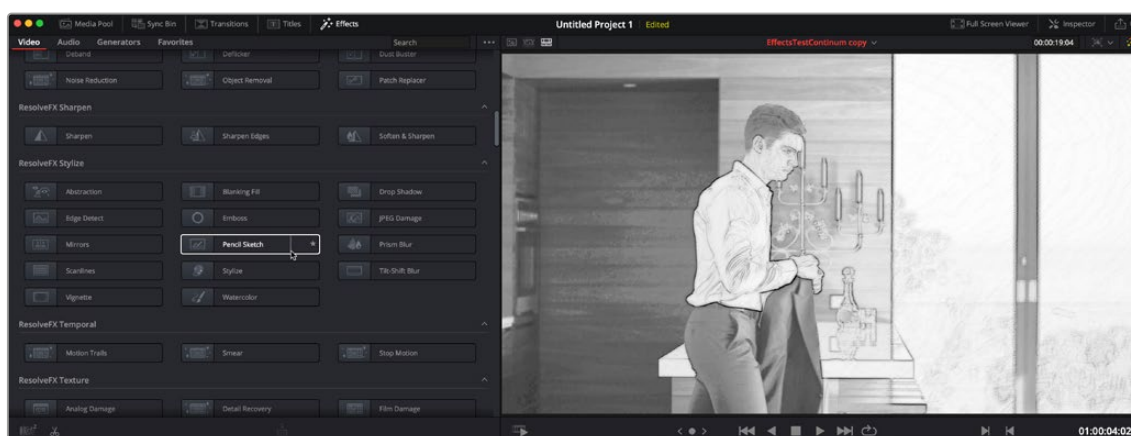
Adding Resolve FX and Other Plugins

The Effects Browser reveals video and audio plugins that you can drag and drop onto to your clips. As far as video plugins go, there are several categories of Resolve FX plugins that accompany DaVinci Resolve, but if you've installed Resolve-compatible OFX plugins, those appear here as well, organized into their own categories. For audio plugins, a set of Fairlight FX plugins comes with DaVinci Resolve, but if you've installed VST plugins on macOS or Windows, or you have Audio Units installed on macOS, those will appear in the Audio category also. When you have a lot of video or audio plugins, a search field makes it easy to find the one you're looking for.

Video Effects

Each available category of video plugins creates different sorts of image effects, such as blurs, lighting effects, or stylization of different kinds. To preview a video effect before placing it on a clip, ensure that “Hover Scrub Preview” is checked in the Effects option menu, then simply hover your pointer over any thumbnail in the Effects tab and move it across the thumbnail. The effect will preview in the Viewer using its default parameters, and scrub through the clip that is selected in the Timeline. If no clip is selected then it will use the clip currently under the playhead.

To activate a specific video effect on a clip, simply drag the thumbnail of the selected effect to a clip on the Timeline. In the Cut page, you can also double-click the thumbnail to apply the effect to the selected clip. To adjust the effect's parameters, open the Effects tab in the Inspector.



Scrubbing over an Effect Thumbnail previews that effect in the Viewer.

Audio Effects

Audio plugins let you process your audio in different ways, adjusting the tone using equalization, changing dynamics with compression, or adding effects such as echo, reverb, flange, or modulation. To activate a specific audio effect on a clip, simply drag the thumbnail of the selected effect to a clip on the upper or lower Timelines in the Cut page, or onto a clip in an audio track on the Edit page. Once the effect has been dropped on a clip, its audio plugin controls will open for you to adjust the plugin's parameters.

Generators

The various video generators included in DaVinci Resolve can be previewed by hovering your pointer over any thumbnail in the Generators tab. To edit a generator into your Timeline, simply grab the thumbnail of the generator you wish to use, and place it in your Timeline in the Edit page, or in either the upper or lower Timelines in the Cut page.

For more information on using all the included video and audio effects in DaVinci Resolve, see *Chapter 47, “Editing, Adding, and Copying Effects and Filters.”*

Interactive Resolve FX and Fusion Effect Overlays in the Viewer

If you’ve applied a Resolve or Fusion effect that uses on screen controls, you can use these controls in the Cut Page viewer. To do so, open the Tools panel in the Cut page, and select the Effects Overlay icon at the far right.

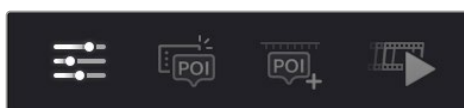
You have two options below to either manipulate the Open FX or Fusion Overlays; choose the appropriate version for your effect. The on screen controls will then appear in the Viewer.



Clicking on the Effects Overlay icon (circled) will activate the on screen controls of Resolve or Fusion effects in the Cut Screen viewer.

Clip Tools

Clicking the Tools button in the Viewer toolbar opens up additional categories of controls that let you apply various effects to the current clip, or adjust whatever effect is already applied to the current clip. Many of the parameters of these effects have corresponding onscreen controls that let you make visual adjustments directly in the Viewer.



The Tools button

Shared Controls

Every category of effect at the bottom of the Viewer has two shared controls. A toggle control at the far left lets you enable or disable an effect without losing whatever custom adjustments you've made. A reset control at the far right lets you reset every parameter within a particular category of controls to the default settings.



The Toggle button (at left) and Reset button (at right)

Transform

When you select Transform, onscreen transform controls appear that let you directly manipulate the image in the Viewer. You can drag anywhere within the clip's bounding box to adjust pan and tilt, drag any diagonal corner to proportionally resize, drag any top/bottom/left/right side to squeeze or stretch width or height, or drag the center handle to rotate.



Onscreen Transform controls in the Viewer

TIP: While dragging a clip using the onscreen controls to reposition it, holding the Shift key down constrains movement to just the X or Y axes.

The onscreen controls also correspond to the following editable parameters, which are also editable in the Video Inspector and in the Edit Sizing mode of the Sizing palette in the Color page:

Zoom Width and Height: Allows you to blow the image up or shrink it down. The X and Y parameters can be linked to lock the aspect ratio of the image, or released to stretch or squeeze the image in one direction only.

Position X and Y: Moves the image within the frame, allowing pan and scan adjustments to be made. X moves the image left or right, and Y moves the image up or down.

Rotation Angle: Rotates the image around the anchor point.

Pitch: Rotates the image toward or away from the camera along an axis running through the center of the image, from left to right. Positive values push the top of the image away and bring the bottom of the image forward. Negative values bring the top of the image forward and push the bottom of the image away. Higher values stretch the image more extremely.

Yaw: Rotates the image toward or away from the camera along an axis running through the center of the image from top to bottom. Positive values bring the left of the image forward and push the right of the image away. Negative values push the left of the image away and bring the right of the image forward. Higher values stretch the image more extremely.

Flip Image: Two buttons let you flip the image in different dimensions.

- **Flip Horizontal control:** Reverses the image along the X axis, left to right.
- **Flip Vertical control:** Reverses the clip along the Y axis, turning it upside down.

Crop

The Cut page has a set of onscreen controls you can use to directly crop the image in the Viewer. Each side of the image has an individual handle for cropping just that side. These parameters are also editable in the Video Inspector and the Color page Sizing palette.



Onscreen Crop controls in the Viewer

The Crop effects also correspond to an additional set of cropping parameters, with an additional control for softness:

Crop Left, Right, Top, and Bottom: Lets you cut off, in pixels, the four sides of the image. Cropping a clip creates transparency, so whatever is underneath shows through.

Softness: Lets you blur the edges of a crop. Setting this to a negative value softens the edges inside of the crop box, while setting this to a positive value softens the edges outside of the crop box.

Dynamic Zoom

The Dynamic Zoom controls, which are off by default, make it fast and easy to do pan and scan effects to zoom into or out of a clip. A set of two onscreen controls let you create a Dynamic Zoom effect. A green box shows the starting size and position of the animated transform, while a red box shows the ending size and position of the animated transform. Drag anywhere within either bounding box to reposition either the start or the end of the animated effect, and drag any of the corners to adjust the size at the start or end. A motion path appears to show you motion that's being created. Adjusting the Dynamic Zoom controls automatically enables dynamic zoom. These controls are also available in the Video Inspector.



Dynamic Zoom controls in the Viewer

These controls correspond to two parameters in the toolbar (Dynamic Zoom is also editable in the Video Inspector).

Zoom/Pan/Angle Presets: Let you enable or disable preset positions for the zoom level, pan location, and angle of this effect.

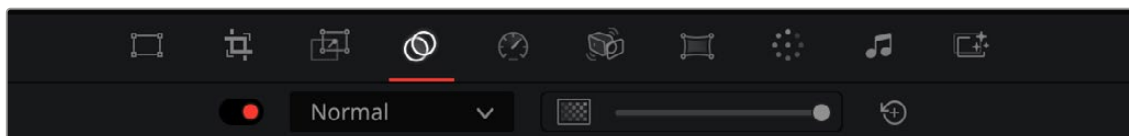
Swap: This button reverses the start and end transforms that create the dynamic zoom effect.

Ease Buttons: Lets you choose how the motion created by these controls accelerates. You can choose from Linear, Ease In, Ease Out, and Ease In and Out.

TIP: While dragging dynamic zoom outlines to reposition them, holding the Shift key down constrains movement to just the X- or Y-axis.

Composite

Two controls let you create transparency and use composite modes to create different compositing effects (also called Blend modes or Transfer modes). These controls also editable in the Video Inspector.



Composite controls in the Viewer

Composite Modes: Composite modes blend two superimposed clips together on the Timeline using different kinds of math to achieve differing results, to create transparency effects, increase image exposure, and combine multiple clips into a single image in many creative and useful ways. All Composite modes interact with the Opacity slider.

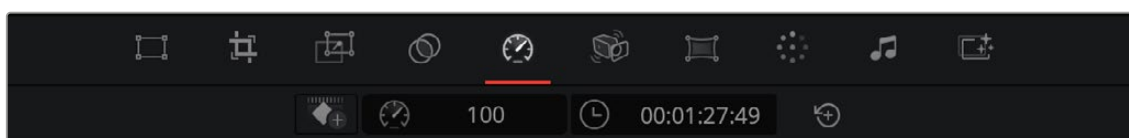
For more information on what each Composite mode does, see *Chapter 50, “Compositing and Transforms in the Timeline.”*

Opacity: This slider lets you make a clip more transparent, over a range from 0 (totally transparent) to 100 (totally opaque). When set to a value less than 100, the selected clip is mixed with whatever video clip is underneath it on the Timeline, using the Composite mode that’s currently selected. If no clip appears underneath the Timeline, then the clip is mixed with black and will work similarly to a fade.

TIP: If a superimposed video or still image clip in the Timeline has an alpha channel, that alpha channel automatically creates transparency within that clip, compositing it against whatever is in the track underneath. There’s no need for you to do anything for this to work.

Speed

Speed effects let you speed up, slow down, or otherwise change the playback speed of clips in the Timeline. When you change the speed of a clip, that clip’s duration also changes to reflect a shorter clip that plays faster, or a longer clip that plays more slowly. Speed effects change both video and audio playback, but the audio of sped up or slowed down clips is always pitch corrected. Speed effects applied in the Cut page also appear and are editable via several different methods in the Edit page timeline and the Video inspector.



Speed Effects controls in the Viewer

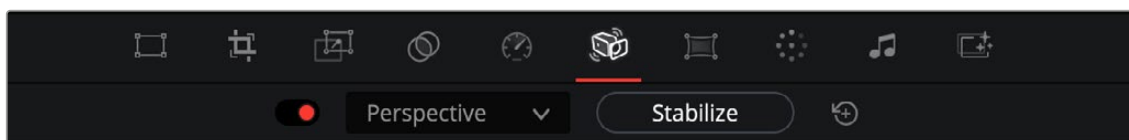
Add Speed Point: Lets you add a speed change point at the playhead position for speed ramping effects.

Speed: Changing this value lets you speed up or slow down playback by a simple numeric multiplier. You also have the option of choosing a negative value to create reverse speeds.

Duration: When you retime a clip, the Duration field lets you see how the change you're making affects the new duration based on the original duration of the clip with no speed effect applied.

Stabilization

The Image Stabilization controls use warping and/or translation to let you smooth out or even lock unwanted camera motion within a clip. The analysis is performed in such a way as to preserve the motion of individual subjects within the frame, as well as the overall direction of desirable camera motion, while correcting for unsteadiness.



Stabilization controls in the Viewer

To stabilize an image, all you need to do is to choose a Stabilization Method from the drop-down (see below for more information), and then click the Stabilize button. DaVinci Resolve analyzes the current clip, and applies a stabilization effect.

The rest of the stabilization controls let you refine the result. Whenever you adjust any of these parameters, you must click the Stabilize button again for the effect to be updated.

Stabilization Method: A drop-down menu provides three different options that determine how the selected clip is analyzed and transformed during stabilization. You must choose an option first, before clicking the Stabilize button, because the option you choose changes how the image analysis is performed. If you choose another option, you click the Stabilize button again to reanalyze the clip.

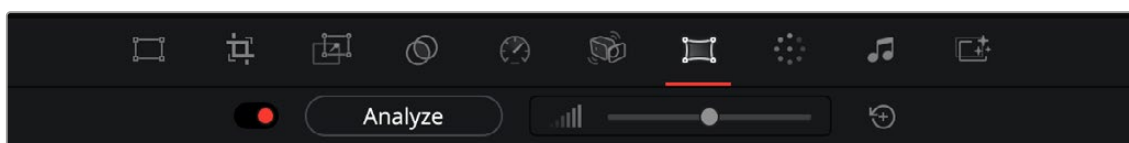
- **Perspective:** Enables perspective, pan, tilt, zoom, and rotation analysis and stabilization.
- **Similarity:** Enables pan, tilt, zoom, and rotation analysis and stabilization, for instances where perspective analysis results in unwanted motion artifacts.
- **Translation:** Enables pan and tilt analysis and stabilization only, for instances where only X and Y stabilization gives you acceptable results.

Stabilize: Clicking this button on a previously unstabilized clip analyzes the motion in that clip and applies an initial smoothing effect. Clicking this button on a clip that's already been analyzed lets you recalculate a modified stabilization effect.

TIP: These controls are identical to those found in the Video Inspector and the Color page Tracker palette, and populate the same image processing data. This means that you can use the Stabilization found on the Cut page, and then use the stabilization graph and controls found in the Color page to refine the results, if necessary.

Lens Correction

Lens Correction presents two controls that let you either correct lens distortion in the image, or add lens distortion of your own for effect. These controls are also editable in the Video Inspector and Color page Edit Sizing palette.



Lens Correction controls in the Viewer

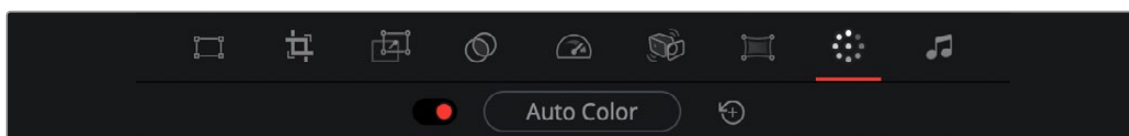
Analyze: Automatically analyzes the frame in the Timeline at the position of the playhead for edges that are being distorted by wide angle lens. Clicking the Analyze button moves the Distortion slider to provide an automatic correction. If you're analyzing a particularly challenging clip, a progress bar will appear to let you know how long this will take.

Distortion: Dragging this slider to the right lets you manually apply a warp to the image that lets you straighten the bent areas of the picture that can be caused by wide angle lenses. If you clicked the Analyze button and the result was an overcorrection, then dragging this slider to the left lets you back off of the automatic adjustment until the image looks correct.

Color

The Color section of the Tools consists of only one option: Auto Color. The Auto Color command provides a quick way to automatically balance the blacks and whites of a clip based on the current frame at the position of the playhead. Using advanced algorithms, based on the DaVinci Neural Engine, it provides superior results when automatically adjusting color balance and contrast.

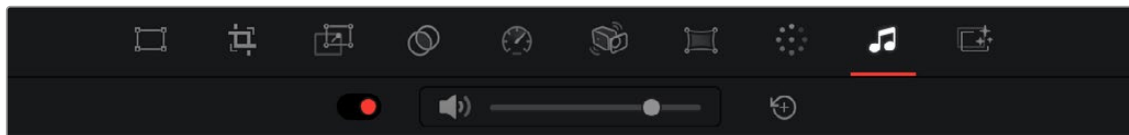
For more details on using Auto Color, see *Chapter 127, "Automated Grading Commands and Imported Grades."*



Color controls in the Viewer

Audio

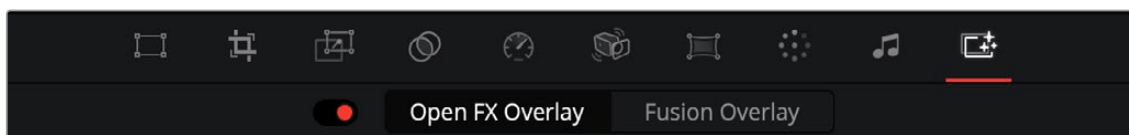
A slider lets you adjust audio levels of the current clip in the Viewer, making the volume of that audio clips softer or louder. This is identical to each clip's volume setting in the Edit and Fairlight pages.



Audio controls in the Viewer

Effects Overlay

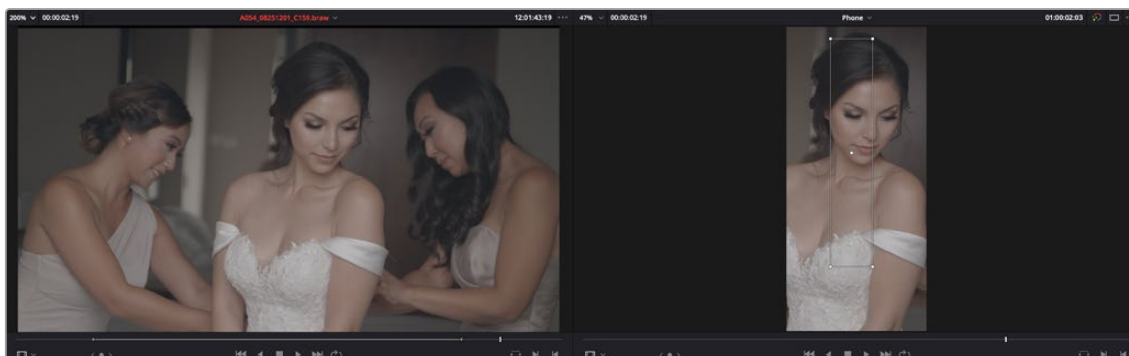
Turning on either Open FX or Fusion overlays lets you use the on screen viewer controls for those effects that use them.



The Effects overlay activates the on screen viewer controls for effects.

Smart Reframe (Studio Version Only)

The Smart Reframe feature in DaVinci Resolve makes it easier to quickly reframe material across extreme aspect ratio changes. It's useful for situations where you've shot a 16:9 horizontal video and find yourself needing to create a vertically-oriented 9:16 version for mobile phones and social media deliverables or using 4:3 archival footage in a 2.39:1 widescreen movie. Smart Reframe can be used manually or automatically executed using the DaVinci Resolve Neural Engine.

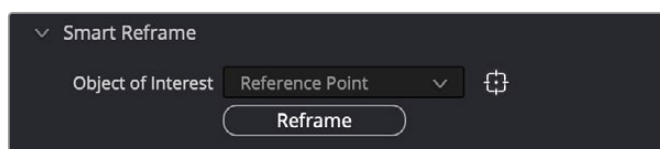


Smart Reframe in action with the Reference Point bounding box active (right)

The Smart Reframe tool is found in the Sizing tab of the Inspector and is available in both the Cut and Edit pages.

To use the Smart Reframe tool:

- 1 Duplicate your Timeline, right-click the Timeline and choose Timelines > Timeline Settings, and click Use Custom Settings to change the Timeline Resolution to the aspect ratio needed for delivery. Make sure that “Mismatched resolution files” is set to “Scale full frame with crop,” and click OK.
- 2 Select one or more clips you want to reframe, and open the Inspector to the Video tab.
- 3 Open the Smart Reframe controls, leave the Object of Interest drop-down menu set to Auto (if you’ve selected more than one clip, Auto is the only setting available), and click “Reframe.” DaVinci Resolve will analyze your footage and should automatically adjust each individual clip’s position to a more aesthetically pleasing framing.
- 4 (Optional) If the “Auto” setting does not give you desirable results for a particular clip, you can manually select the main subject using the following steps.
 - a) To manually select the subject area, choose “Reference Point” from the Object of Interest drop-down menu, and click the Target icon just to the right of the menu. This automatically sets the Viewer mode to Smart Reframe, exposing the onscreen controls for choosing a reference.
 - b) Drag the Reference Point bounding box around the main subject of interest in the frame. You may use the Transform controls directly above in the Inspector to move the source clip around if your subject is outside the current framing.
 - c) Click “Reframe.”

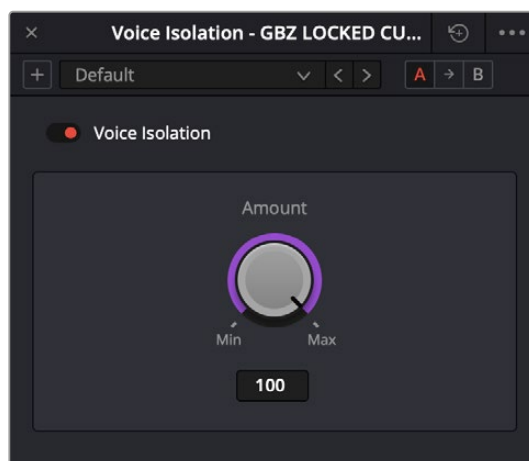


The Inspector’s Smart Reframe controls showing the manual reference point selected

DaVinci Resolve locks onto and, if necessary, tracks your subject using the reference you’ve selected, automatically panning and scanning the original clip as needed to keep the reference within the new aspect ratio. While involving a bit of manual adjustment, this function still dramatically reduces the time involved in pan and scanning footage by manually adjusting and keyframing the sizing controls.

Voice Isolation (Studio Version Only)

Voice Isolation uses an AI model trained for any type of human voice, male or female, young or old, which lets you completely remove loud, undesirable sounds from existing voice recordings with incredible results.



The Voice Isolation plugin

You can get rid of all kinds of background noise, from air conditioners or fans to jackhammers, restaurant background noise, music playing during a featured piece of dialogue, and more.

The Amount control adjusts the mix between the original source and the isolated voice. When set to 50, the mix is roughly equal. Values between 70 and 80 work well for natural results while strongly isolating the source.

Although Voice Isolation can be used on an audio track in real time, it cannot be used on live audio input. It can be used on any mono, stereo, or dual-mono audio track, but it isn't supported on tracks with more than two channels.

Dialogue Leveler

The Dialogue Leveler analyzes source material to detect dialogue and then “rides down” louder areas, “lifts up” softer areas, and lowers background sounds that are not dialogue.

It works without the typical “pumping” or other unwanted side effects of dynamics processors (compression/limiting) to produce results like those achieved through detailed, manual clip gain adjustments or “riding” the track with fader automation.

The Dialogue Leveler can often “fix and mix” a recording made with a single boom mic, where one character is closer than another or turns away from the mic. and levels it out so that the original relative dynamics are preserved while increasing intelligibility and average level.



The Dialogue Leveler plugin

Although Dialogue Leveler can be used on an audio track in real time, it cannot be used on live audio input. It can be used on any mono, stereo, or dual-mono audio track, but it isn't supported on tracks with more than two channels.

Dialogue Leveler controls

Dialogue Leveler: Enables or disables the plugin.

Waveform Processing display: This shows a scrolling waveform and a gray line indicating Dialogue Leveler processing. However, this display is not available when using the Dialogue Leveler found in the Inspector to process audio clips on the timeline.

Preset Menu: This menu provides the following options:

- **Allow wider dynamics:** This is the default and is best for audio with a wider-ranging dynamic from loud to soft, and where the clip levels are medium to high.
- **More lift for low levels:** Select this option if the source has more low-level dialogue that you want to boost.
- **Lift soft whispery sources:** Select this option if the source has whispered dialogue and background noise.
- **Optimize moderate levels:** Select this option if the source is at medium levels throughout.

Reduce Loud Dialogue: When enabled, louder dialogue is ridden downward when it peaks, acting like a “perfect limiter” where you don’t have to adjust the threshold or time parameters. Due to the “near real-time” functionality, analysis occurs prior to audible playback for optimal results.

Lift Soft Dialogue: This setting finds and lifts low-level dialogue, evening out material that is quieter while more variable in level. Because the process is dialogue-focused, it doesn’t tend to raise background sounds unless they’re audible at the same time as the dialogue.

The Lift Soft Dialogue option is often the most useful of the three options, as it makes quieter lines of dialogue louder and naturally smooth without boosting background noise.

Background Reduction: This option gently removes background noise based on the internal preset selected in the Preset menu.

Output Gain: You can use this control to adjust the output level by clicking and dragging it or entering a value in the numeric field. The range of the Output Gain control of 0.0 to +6.0 dB.

TIP: Before using the Dialogue Leveler on a track, it can be useful to set an optimum baseline level for use with the default “Allow wider dynamics” preset by normalizing the audio.

For more information on normalizing audio, see *Chapter 176, “The Fairlight Inspector.”*

Music Remixer (Studio Version Only)

This DaVinci Neural Engine AI-based plugin splits music into individual stems: Vocals, Drums, Bass, Guitar, and “Other” (“everything else”), letting you creatively rebalance or remix the track.

You can use the level controls to adjust the volume of each stem and the Mute buttons to remove or bring them back into your mix. For example, you may want to remove or lower a vocal because it clashes with a featured piece of dialogue.



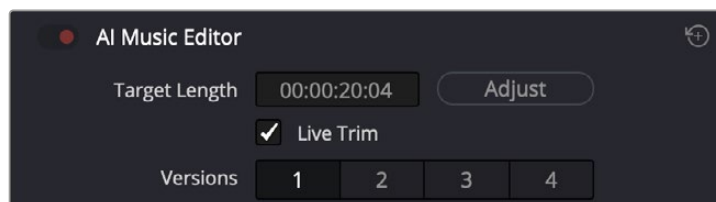
The Music Remixer plugin

Music Editor (Studio Version Only)

DaVinci Resolve 20 supports seamlessly extending or shortening music to the length of your edit. The new AI powered Music Editor can be accessed in the Audio tab of the Inspector. You can enter a desired duration and press the Adjust button in order to shorten or extend the clip to that duration.

Note: Music Editor is an “intelligent edit” that works effectively with beat-driven music (pop, dance, etc.). It is not intended for free form, ambient or non-beat driven material.

Music Editor does not use time compression/expansion or alter the pitch of the music, but instead uses editing techniques like a music editor might to fit the material to desired length to picture. It will look for logical places to create transitions based on whether you’re extending length (by repeating sections), or removing material, and attempt to use the ending or fade of the original composition if it times well. If the ending is abrupt, it’s up to you to manually create a fade out.



The Music Editor in the Audio Inspector

To Use Music Editor to retime a music clip:

- 1 Select the clip on the timeline you wish to retime. This clip does need to be beat-driven music.
- 2 In the Audio Inspector under AI Music Editor, enter a new Target Length for the music duration you want.
- 3 Press the Adjust button.
- 4 Alternatively, you can check the Live Trim box to use the Trim tools to resize the clip and dynamically retime the music on the timeline. You can turn the checkbox off to return to normal editing (e.g., trimming) for that clip after the desired edit has been created.

Once the analysis is complete, four versions of potential edits you can use appear at the bottom, each with a different approach to the edit, numbered 1-4. Any subsequent editing can be done almost instantaneously. The edited pieces of music show indicators on the clip to mark the portions where cuts and transitions occur.



The initial music is too long



The Music Editor has retimed the audio clip to match the length of the video. Note the jagged lines on the audio clip indicating where the original was cut. In the waveform you can see the end of the music is retained in the retimed clip.

TIP: While timings can be quite close, sometimes they may not be exact. If you need to get an exact timing after working with AI Music Editor, you might use the Elastic Wave tool on the Fairlight page and choose the algorithm for general purpose (music).

The edited clip is actually a specialized Music Editor clip which allows you to further trim your edit. If you want to access the underlying edit, you can also right-click on the clip and decompose, exposing the edit points and any crossfades created. However, keep in mind that once decomposed, you can't trim the length of the clip as an entity as each editing snippet is now a separate clip.

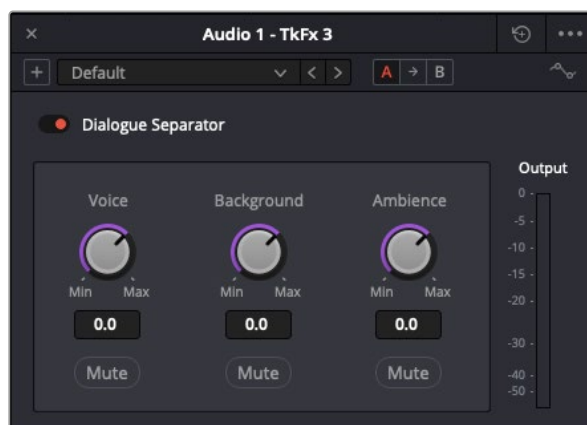
Dialogue Separator Track FX (Studio Version Only)

Dialogue Separator uses DaVinci Neural Engine AI to give you individual control over the level of dialogue, background sounds, and the reverberant field or ambient room sound ("ambience").

The Dialogue Separator is useful when, for example, a source file has a great roomy sound, but there's music or crowd noise in the background. In this instance, you can rebalance the audio to reduce the background sounds or adjust the room sound to make the dialogue more intelligible.

Another use case would be when someone moves toward the camera in a roomy environment while speaking. You could use automation to rebalance the audio so that the ambient room sound gradually becomes "drier" as they get closer, without affecting the background.

Dialogue Separator is currently a mono-only plugin. If your source is stereo, you'll need to split it into dual-mono and process both mono tracks separately.



The Dialogue Separator plugin

Ducker Track FX

This plugin lets you use the audible signal from a track or multiple tracks to lower the level of another.

Although it can be used in many creative ways, a common use of “ducking” is automatically lowering a music or sound effects bed so that dialogue or a VO track can be heard more prominently in a mix.

This is achieved without compressing the incoming signals. When set up correctly in the Ducker interface or via the Audio tab in the Inspector, your audio tracks can practically “mix themselves.”

In a simple use case, the Ducker uses level changes of a single incoming dialogue or VO track to decrease the volume of the sound effects bed of a noisy freeway.

In a more complex situation, such as a scene where three characters are conversing next to the same noisy freeway, a single Ducker plugin can use the level changes of each dialogue track to attenuate the traffic noise.

With only one input source (Dialogue 1), the Ducker would only be effective when the first character speaks. However, this new feature lets you add the other two dialogue tracks (Dialogue 2 and Dialogue 3) as input sources for attenuating the traffic noise.



The Ducker plugin

Using the Ducker

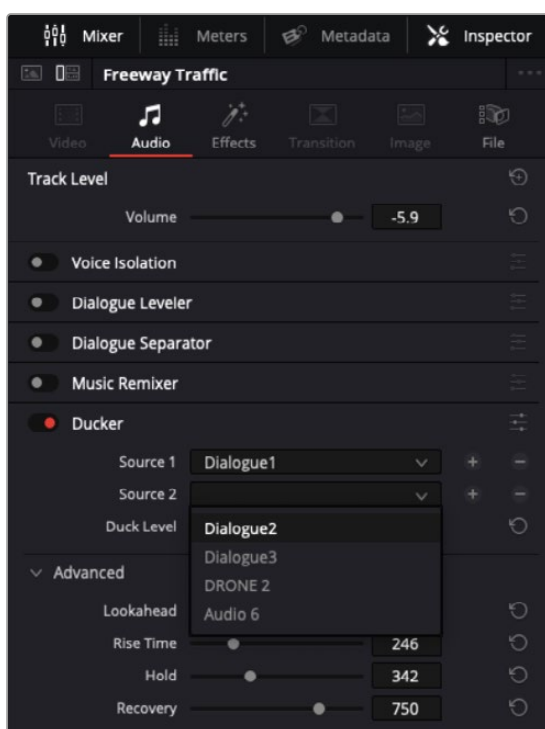
Continuing with the use cases mentioned above:

- 1 Choose the target track that you want to attenuate (the Freeway Traffic track), and open the Ducker.
- 2 Click the Source dropdown and select Dialogue 1 as the input source.
 - To use additional input sources, Command-click the other track names; in this example, Dialogue2 and Dialogue3.
 - Play a bit of your sequence, and adjust the duck level and other controls to taste.

Adding Input Sources via the Inspector

If you prefer working within the Inspector:

- 1 Select the Freeway Traffic track, ensure the Ducker is switched on, and open the Inspector's Audio tab.
- 2 Click the plus sign to the right of the Source 1 dropdown containing Dialogue1 to create a dropdown menu for Source 2, then choose Dialogue2 from the list.
- 3 Click the plus sign to the right of the Source 2 dropdown to create a dropdown menu for Source 3, then choose Dialogue3 from the list.



Adding Ducker input sources in the Inspector

Ducker Controls

- **Duck Amount:** This control determines level attenuation on the target track in dB with a default value of 2.7 dB. Most of the time, a value between 2.0 dB and 5.0 dB works best.
- **Lookahead:** Adjusts the amount of lookahead time in milliseconds, with a default value of 15ms. Lookahead allows the Ducker to start attenuating the destination track before the source track is audible.

The larger the value, the more the Ducker will “pre-anticipate” the source track. If you want the Ducker to start lowering the sound before a line of dialogue or VO begins, you can experiment with higher values, but the default will usually produce good results.

- **Rise Time:** This determines how many milliseconds it takes for the Ducker to reach the amount of reduction specified by the Duck Amount. The default value is 10 milliseconds, which should work for most situations. Longer Rise Times may be useful with higher Duck Amount values and longer gaps in the source track.
- **Hold:** This parameter determines how long the target signal will be attenuated. The default is 150ms.
- **Recovery Time:** This value indicates how quickly the target signal will return to its previous level after attenuation. The default time value is 750ms, but setting the correct Recovery time is crucial to a natural-sounding ducking process, so you should try different values to find the best results.

Ducker Graph Display

This area shows a gray line representing the incoming signal against the action of the Ducker, shown in yellow. Dips in the yellow line signify attenuation of the target track based on the Duck Amount in combination with the other parameters.

Beat Detector (Studio Version Only)

DaVinci Resolve can automatically detect beats for a piece of music on the timeline, and supports snapping to these beat marker locations. This can be especially useful when trying to make edits to the beat.

NOTE: The Beat Detector will only work effectively with beat-driven music, and with 4/4 or 3/4 time signatures at this time.

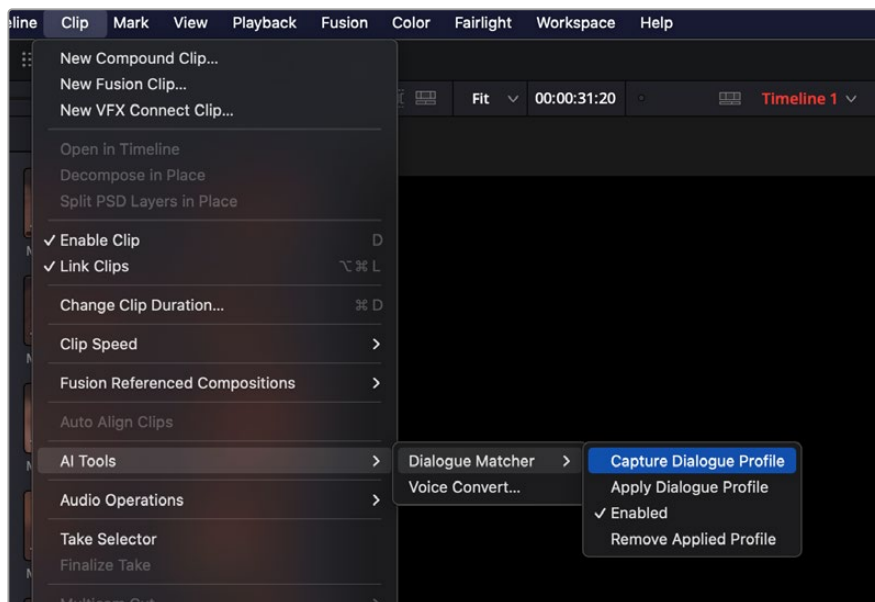
To detect beats, right-click on the timeline audio clip and select Detect Music Beats from the context menu. Once analyzed, the timeline clip displays beat indicator overlays that you can snap other timeline elements to.



A music track with the Beat Detector applied. The lined overlays match up with the beat of the music.

Dialogue Matcher (Studio Version Only)

Dialogue Matcher allows you to match one piece of dialogue to another, matching level, EQ, and reverb. This can help with situations where new recordings may not match the environment of existing ones. The Dialogue Matcher is a fully automated tool that only requires the user to capture a tone profile from a clip and then to apply it to another.



Capturing a Dialogue Profile from a clip using the main menu

To Use Dialogue Matcher:

- 1 Select a reference clip whose dialogue you want to match, right-click, and choose Clip > AI Tools > Dialogue Matcher > Capture Dialogue Profile.
- 2 Select the clip whose dialogue you want to change, right-click and choose Clip > AI Tools > Dialogue Matcher > Apply Dialogue Profile.
- 3 You can enable or disable the Dialogue Matcher on a clip by right-clicking and choosing Clip > AI Tools > Dialogue Matcher > Enable.
- 4 You can Remove the Dialogue Matcher on a clip by right-clicking and choosing Clip > AI Tools > Dialogue Matcher > Remove Applied Profile.

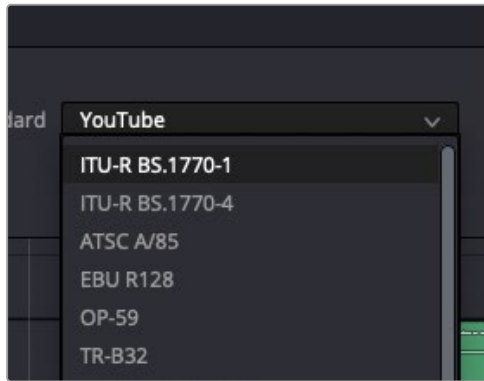
Audio Assistant (Studio Version Only)

With the Audio Assistant, DaVinci Resolve can automatically organize and color code your tracks, even out your dialogue levels, and adjust the mixer faders to create a professional-quality mix of your music, sound effects, and dialogue.

During this process, your dialogue will also be cleaned up, and if needed, Voice Isolation and De-Essing will be applied. Ducking will be applied to music tracks, any required automation will be written, and the appropriate mastering and finalizing plugins will be added and adjusted to help achieve the correct loudness level for your chosen delivery standard.

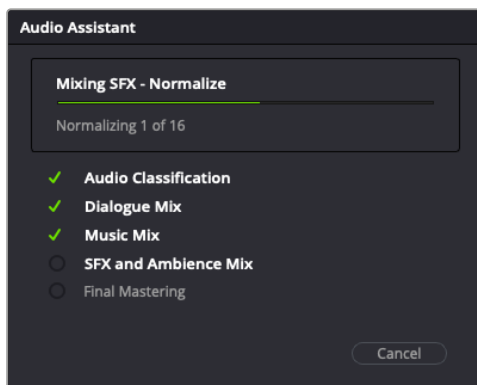
To use AI Audio Assistant:

- 1 Open a timeline containing unmixed audio. The initial fader levels really don't matter as Audio Assistant will handle all fader adjustments during the mixing process.
- 2 To begin the automatic mixing process, choose Timeline > AI Tools > Audio Assistant from the Timeline menu, select the Delivery Standard for your mix from the dropdown menu, and click the Auto Mix button.



Audio Assistant dialog - Delivery Standard

- 3 The Audio Assistant dialog changes to display the progress of the mixing process with an option to cancel it.



Audio Assistant dialog - Mix Progress

- 4 When the mixing process has concluded, you can play back the mix and, if needed, make further adjustments or completely undo it.

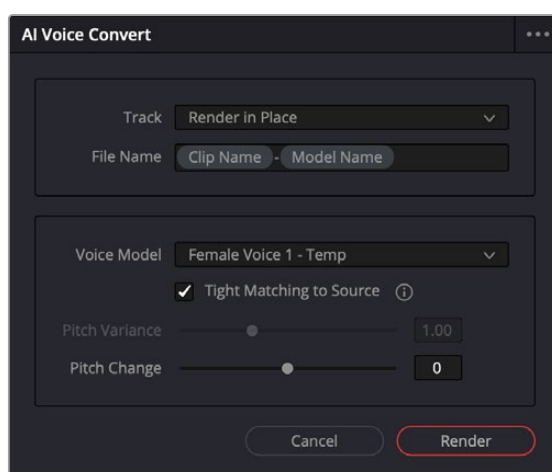
NOTE: If a track is in the wrong category, right-click the corresponding track header and choose the correct category from the Track Category section of the contextual menu, and then rerun the Audio Assistant.

Voice Convert (Studio Version Only)

DaVinci Resolve has an incredibly sophisticated AI-driven voice conversion tool that allows you to essentially swap one speaking voice for another. The use cases for this tool are wide reaching and even somewhat controversial, however this Voice Convert toolset can be an invaluable resource in a post production. It allows you to take a generated voice model that has been captured by DaVinci Resolve and “drive” that voice model using another voice recording as its foundation.

In simple cases, you can replace your own voiceover tracks with a voice that has more depth and gravitas to it, or even switch the gender of your own voice if you feel it’s more appropriate for the subject matter. Perhaps you recorded audio in a noisy environment and wanted to make a pristine version instead, by using your own voice as a model, to replace your original noisy tracks. In more complicated scenarios you can train your own voice convert model on your actor’s voice patterns and use that to change and replace their previous voiceover, without having to get them back into the recording booth. Finally, it is even possible to change the voice of an actor completely to another voice all while staying perfectly in sync with the picture. The inflection, pitch variations, and emotion of the original all remain; it just sounds like someone else.

Needless to say, using this powerful tool requires a certain level of responsibility on the part of the user. Privacy, copyright, and union laws and rules are all in flux at the moment around AI-generated content, and it’s necessary to carefully navigate the ethics involved in its professional use.



The Voice Convert tool

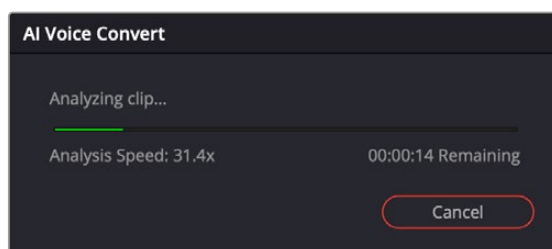
NOTE: Using Voice Convert is very processor intensive and requires your system to have a minimum of 8 GB of video RAM before it will activate, to avoid crashes and system instability. Even with a strong system, creating a Voice Model will require prolonged intensive CPU and GPU power, so multi-tasking while it’s working may be challenging.

Using Voice Convert

Like most of DaVinci Resolve’s AI tools, using Voice Convert is very simple, with all of the complex processing involved handled by the voice model.

To use Voice Convert to change one voice to another:

- 1 Place the clip or clips you want to change the voice of in the timeline, and select them.
- 2 Right-click on the clip and select Voice Convert from the context menu, or select Clip > AI Tools > Voice Convert in the main menu.
- 3 Select the track that you want the new audio to appear in (including overwriting the original).
- 4 Select the file name you want the new audio to be named.
- 5 Select the Voice Model you want to use for the new audio (you can make your own custom voice models as explained below) and its parameters.
- 6 Click Render to start the Voice Convert process.



Voice Convert analyzing a clip

At this point the Voice Convert dialog will appear, showing you the progress of the conversion. Depending on the length and number of clips selected, this can take some time. When the process is done, the new audio will appear in the timeline in the exact same place as the original audio (or on the same place in another track if that option was selected). The new audio clips will also appear in the Media Pool.

Simply play back the timeline to audition the new voice. You can change the audio back to the original by right-clicking on a clip and selecting Revert to Original Voice from the context menu.

TIP: While you can drive voices with different accents, Voice Convert generally can't recreate another accent. For example, if you have a voice model that is a speaker with a British accent and drive that voice with an American accent, you get more of an American version of the British accented speaker. But if you speak with a British accent when driving the British voice model, it will work well.

Voice Convert Settings

Each voice conversion requires selecting set up parameters to drive the model.

Track: Select which track you want the new audio to be placed on. The default is Render in Place, which will overwrite the original audio.

File Name: Set the file name for the new audio clip. You can enter a specific name, or use the metadata variables to create a template for file naming. By default the name is ClipName-(Voice Model Name).

Voice Model: Select which voice model you wish to use for the new audio. There are some built-in options that come with DaVinci Resolve, and any custom options you create will also be available here.

Tight Matching to Source: Check this box to tightly follow the pitch, intonation, and volume variance of the source voice. Uncheck this box to have a looser interpretation using the Pitch Variance tool.

Pitch Variance: When the Tight Matching to Source box is unchecked, this tool becomes active. Lower values create a more monotone interpretation, while higher values have more variance in pitch.

Pitch Change: Use this slider to create a deeper (negative values), or higher (positive values) voice based on the model. The values are in semitones and can be adjusted in 100 cents (.1 semitone) increments.

Creating Your Own Voice Models

It is possible to create your own voice models for use with Voice Convert. These can be of your own voice or even someone else's that you have the rights to. This section goes over some best practices to get a clean and usable voice model.

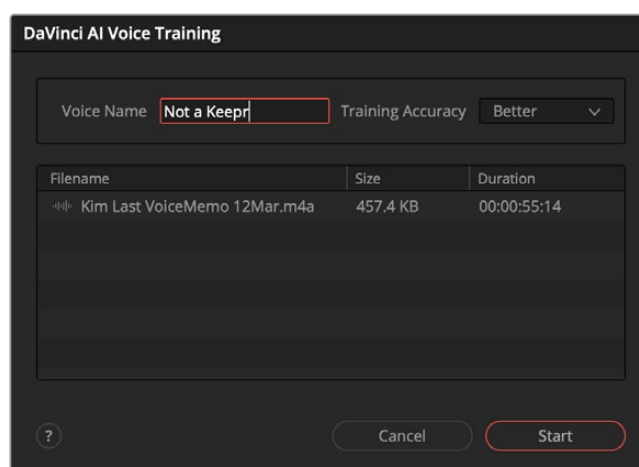
Preparing the Voice Model Assets

You will get the best results using ten minutes or so of clean, high-quality recordings of the voice to be modeled. The audio should be free of excessive ambiance and noise, with little to no dynamics processing. You will want consistent material that sounds natural. It can be useful to have a variety of emotions expressed as well, if you want to hear those results for the driven voice. Treat it like a voice over audition; show your range.

These voice assets do not necessarily need to be in one long clip, though if there are a lot of separate files, it can be helpful to consolidate them all in a single bin for organization. The multiple clips should be of consistent quality, without audio variations in the recorded sources.

Creating the Voice Model

To generate your own voice model, select one or more clips for analysis in the Media Pool, right-click and choose AI Tools > DaVinci AI Tools Voice Training from the context menu. If this is the first time you've run this tool, DaVinci Resolve will open up the Resolve Extras Download Manager to install the required support files.



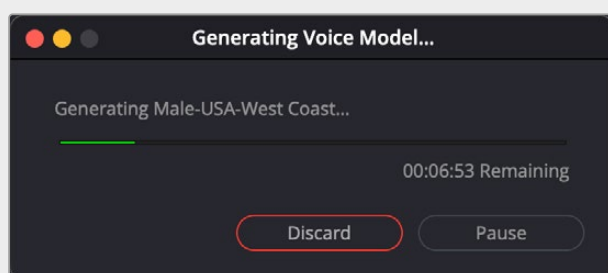
Voice Model Generation

Initially a warning splash screen appears, to ensure you assume full responsibility for how you will use this tool. Once you hit accept, the DaVinci AI Voice Training dialog appears, and there are two options to set.

Voice Model Name: Enter a descriptive name for your new model.

Quality: Choose Faster or Better mode for analysis. Better takes about 3x the time as faster, and 10 minutes or so of material can take a few hours to complete, depending on your computer's power.

Initially creating a voice model launches a foreground process, then once the initial analysis is complete, it will switch to a background process, allowing you to continue to use DaVinci Resolve for other purposes or other apps while it finishes.



Voice Model progress

You can see the progress of the model in the Voice Model Generation Progress indicator in the bottom-right corner, or by selecting Workspace > Voice Model Progress.



The Voice Model Generation Progress indicator will slowly fill blue as the voice model finishes. Hover over the top to get an ETA and progress finished tooltip.

You can also alternatively choose to pause or discard the in-progress model instead. Quitting DaVinci Resolve will also pause the background process, and it will automatically start up again the next time DaVinci Resolve is launched.

Once the model is complete, the new voice preset will appear for use in the Voice Convert Dialog.